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Rock Failure Mechanisms of Flame-Jet Thermal Spallation Drilling—Theory and Experimental Testing

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Certain polycrystalline rocks will fracture into thin, disk-like fragments when exposed to rapid surface heating. Several current hard-rock drilling methods using supersonic flame-jets as heat sources exploit this behaviour for efficient granite quarrying or blasthole formation. The recent application of Weibull's theory of rock failure to quantitatively analyze rock spallation has been extended to allow prediction of chip size distributions and rock surface temperatures at spallation under any intense heat source. Therefore, with measured values of the Weibull distribution parameters, surface spallation temperatures can be estimated theoretically. However, finite-element analysis of intergranular stresses developed during transient heating prior to spall formation indicates that additional microcrack initiation will be favoured thereby altering Weibull parameters measured at room temperature. Nonetheless, experimentally observed spallation characteristics of Barre and Westerly granites are consistent with theory when mechanically determined Weibull parameters are substituted. Spallation temperatures induced by flame-jet heating were estimated to be below 520°C. In all cases, quartz or other mineral solid phase transitions are not expected at these conditions. Laser-induced spallation experiments were less successful, partially because extremely localized heating delivered by the beam greatly complicated the subsequent thermal stress analysis.

NOMENCLATURE

a' = parameter in exponential heat flux distribution;
A = area (m^2), constant in crack size distribution;
C = rock specific heat at constant deformation (J/kg-K);
C_L = chip aspect ratio;
C_p = heat capacity at constant pressure (J/kg-K);
$C_{p,r}$ = rock heat capacity (J/kg-K);
D = diameter of nozzle (m);
E = Young's modulus (Pa);
$G(\sigma)$ = Weibull distribution;
$G_1(\sigma)$ = cumulative spall thickness distribution;
h = convective heat transfer coefficient (W/m^2-K);
I_σ = integral in Weibull strength distribution;
k_{gas} = gas thermal conductivity (W/m-K);
k_r = rock thermal conductivity (W/m-K);
K_{Ic} = critical stress intensity factor (N/m$^{3/2}$);
ΔK_I = stress intensity factor (N/m$^{3/2}$);
l_1, l_2 = crack lengths (m);
l' = initial crack length (m)
m = Weibull homogeneity parameter;
N_i = number of chips liberated per unit area per unit time (1/m^2 sec);
N_{tot} = total number of chips liberated per unit time (1/sec);
Nu = Nusselt number (hD/k);
P = probability of rock failure;
P_λ = total laser power (W);
Q = heat flux (W/m^2), chip distribution leaving the hole;
Q_{max} = maximum or centreline value of heat flux (W/m^2);
Q_r = applied heat flux (W/m^2);
r = radial coordinate;
r' = inside radius in laser heat flux distribution (m);
r'' = outside radius in laser heat flux distribution (m);
r_0 = spreading radius in heat flux distribution (m);
Re_x = Reynolds number based on distance travelled in flat plate boundary layer;
R_h = hole radius (m);
t = time (sec);
t_s = spallation time (sec);
T = temperature (K);
T' = dimensionless temperature;
T_{calc} = calculated temperature (K);
T_{exp} = experimental temperature (K);
T_f = flame temperature (K);
T_0 = reference or initial temperature (K);
T_r = rock temperature (K);
T_{r0} = initial rock temperature (K);
T_s = rock surface temperature (K);
ΔT = temperature change (K);
ΔT_s = rock surface temperature change at spallation (K);
u = displacement (m);
u_i = displacement vector (m);

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u_r = local rock penetration velocity (m/sec);
v = fluid velocity in boundary layer (m/sec);
V = volume (m^3);
V_{dr} = overall rock drilling velocity (m/sec);
x = coordinate normal to flat surface;
α_r = rock thermal diffusivity (m^2/sec);
β_r = rock thermal expansion coefficient (K^{-1});
γ = fracture specific energy (J/m^2);
δ = rock spalling thickness (m);
δ_{bl} = boundary layer thickness (m);
δ_s = thickness of spall (m);
ϵ = strain;
θ, θ_i = inclination angle of boundary;
ν = Poisson's ratio;
ν_{bl} = gas kinematic viscosity (m^2/sec);
$\xi_0 = mV_{dr}x/\alpha_r$;
π = "drillability" ratio;
ρ = density (kg/m^3);
ρ_r = rock density (kg/m^3);
σ = rock thermal stress (Pa);
σ_c = uniaxial compressive rock strength (Pa);
σ_0 = Weibull rock strength (Pa);
σ_s = surface stress (Pa);
σ_t = rock tensile strength (Pa);
μ = shear modulus (Pa);
Φ = value of minimization function;
χ = dimensionless time;
Ψ = crack size distribution function ($1/m^4$);
Ψ' = angle that plane of weakness forms with major stress;
$\omega = mu_r/\alpha_r$ (1/m);
$\omega_1 = mu_r x/\alpha_r$.

INTRODUCTION

The ability to access resources that are several km below the earth's surface routinely or to form large-diameter shafts and tunnels inexpensively ultimately depends on the efficiency of imparting energy to disintegrate or comminute rock. Most often, mechanical or thermal energy delivered to the rock working face produces stresses which accumulate to the point where brittle rock failure is inevitable. Conventional methods for drilling and tunneling have relied exclusively on mechanical rock breakage, often with compressively loaded bits supplying the required stress.

For deep drilling, the primary disadvantages of mechanical methods are bit and drill pipe wear and the large capital expense associated with the drilling rig and ancillary equipment. When hard, crystalline, basement rocks such as granite are encountered, mechanical crushing and grinding of rock are very arduous. This leads to rapid bit wear and frequent bit replacements, which require removal of all drill pipe within the hole and consume substantial rig time. Very low instantaneous penetration rates (1–7 m/hr) in hard rock further lengthen rig time [1]. Although the use of advanced hardened materials, such as synthetic diamonds and tungsten carbide, has significantly improved bit durability and penetration speeds, rotary drilling in hard rock remains an inefficient system prone to rapid wear of all hardware components.

Thermal rock destruction methods use high heat fluxes to induce thermal stresses leading to rock failure. Typically, they rely on radiative sources, such as lasers or electron beams, or convective heating from flame- or plasma-jets [2]. The characteristics of these power sources eliminate the need for initimate contact between rock and machinery. However, in drilling, disintegration of rock should produce rock particles that are easily transported vertically out of the hole by a circulating drilling fluid. Conventional rotary systems have a distinct advantage in this regard, because drilling fluid, often water or a water-based mud, ejected from the bit face serves not only to cool the bit face but also to clean debris away, thereby exposing underlying, unfractured rock to the bit.

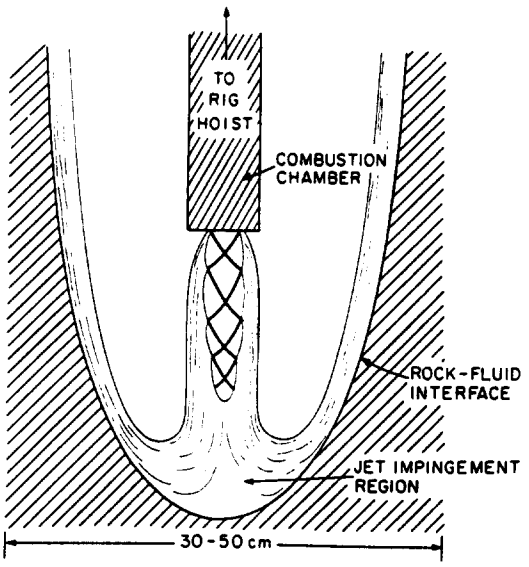


Fig. 1. Schematic of supersonic flame-jet issuing from the combustion chamber nozzle, expanding and impinging on underlying rock.

Thus, for thermal drilling techniques to be commercially competitive, both an increased penetration rate, which collaterally implies rapid cuttings removal and high delivered heat fluxes, and infrequent cutting tool replacements are required. Although radiative or electrical power supplies can readily supply enough energy to penetrate a rock working face by melting or spalling, they provide no means of clearing the rock rubble as it is produced. By contrast, a flame-jet torch can deliver the necessary heat flux and gas jet velocities to spall and clear rock chips while drilling holes [3].

Flame-jet spallation drilling or jet piercing employs a hot jet of combusted fuel, usually flowing at supersonic speeds and directed at underlying rock (Fig. 1). As with most other thermal methods, high surface heating rates (0.5–10 MW/m^2) induce rapid accumulation of thermal stress within the rock, causing it to fracture or spall into fragments often smaller than a single grain (< 1 mm). Occasionally, if the heat flux is low, relatively thick (about 0.5–1 cm), plate-like spalls can be generated. Failure mechanisms leading to spall formation have been a controversial subject for many years, and the literature contains several contradictory theories. A discussion of these and a new theoretical treatment of spallation mechanics will be presented in this paper.

REVIEW OF PREVIOUS FIELD EXPERIMENTS

Despite uncertainty about the most appropriate description of thermoelastic behaviour of rocks under

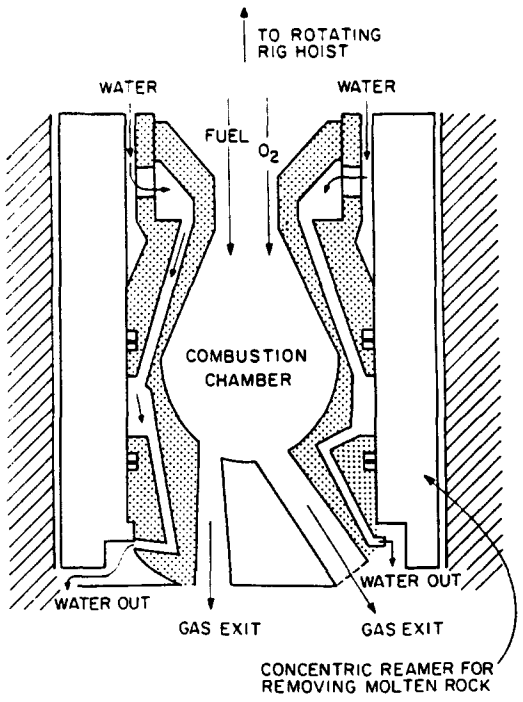


Fig. 2. Schematic of typical Linde rotating, two-port burner and reamer assembly [4].

these conditions, the high costs associated with rotary or percussive drilling in hard rocks have prompted field applications of spallation to taconite mining [4], granite quarrying [5] and well formation [6]. Since the 1940s flame-jet spallation of hard, polycrystalline rocks has been exploited commercially by the Linde Division of Union Carbide. Taconite (iron-ore) surface mining techniques rely on formation of several blastholes, usually less than 20 m deep, that are packed with high explosives and then detonated to disintegrate the ore. Blasthole drilling with fuel oil–oxygen flame-jet tools manufactured by Linde Division proved more economical than mechanical drilling, and thousands of feet of hole have been drilled with the Linde process.

Browning *et al.* [5] demonstrated that air was a more economical choice of oxidant. Furthermore, for most readily spallable rocks, one directed air-fired flame-jet exiting through an on-axis converging–diverging nozzle was sufficient, in contrast to the Linde two-port design (Fig. 2). As in the Linde system, Browning's supersonic flame-jet rapidly spalled the rock surface with the gaseous combustion products removing the chips and debris away from the fresh rock surface and lifting them out of the hole. Using a burner that combusted compressed air and fuel oil at about 3.4 MPa (500 psia), Browning was able to drill a 20–25-cm dia hole at an average rate of 16 m/hr (52 ft/hr) in Conway, NH, granite (Table 1). Near the end of the 335 m (1100 ft) hole, once the drill had passed through most of the pre-existing fractures, the instantaneous penetration speed rose quickly to over 30 m/hr.

Browning's next field operation at the Rock of Ages quarry in Barre, Vermont, concentrated on creating a larger diameter well. The utility flows were identical to those in the Conway experiment, but the lower chamber combustion pressure [0.9 MPa (125 psia)] could only produce penetration rates of 6–9 m/hr. A potential limitation of spallation was identified during the drilling exercise at Barre. The airflow [34 m^3/min (1200 scfm)] delivered to the burner was simply unable to effectively lift rock chips the total hole length (130 m). Future field drilling tests will demand greater compressor capacity to provide sufficient lifting capability.

Investigators at Los Alamos National Laboratory recognized that flame-jet spallation drilling could markedly reduce drilling costs associated with development of hot dry rock geothermal reservoirs in granite formations [1]. The Los Alamos team modified Browning's umbilical hose system and tested it in the field [7]. To permit ignition with the drilling apparatus already downhole without retrieving the burner, a spark plug was inserted near the fuel oil entry into the combustion chamber (Fig. 3). Utility flows matched those used by Browning in his Barre campaign, thereby producing almost identical hole characteristics (Table 1).

Although previous field tests and model studies [8] of flame-jet spallation drilling have repeatedly confirmed technical process feasibility, little well-defined experimentation or quantitative analysis of rock spallation mechanics has been completed. Rocks spall under high heat flux and drilling can be achieved by exploiting this behaviour, but the fundamentals of rock failure under these conditions are not well understood. Experiments under controlled conditions, such as those performed as a part of our research, are still needed to provide a better characterization of important rock failure mechanisms.

REVIEW OF EARLIER THEORIES OF ROCK SPALLATION

During the past 50–60 yr, the development of a mechanistic representation of rock spallation has been

Table 1. Summary of spallation drilling experience in crystalline rock

Researcher	Location	Depth	Rate	Diameter	Compressor pressure	Air flow
Browning	Conway, NH	335 m (1100 ft)	15.8 m/hr (51.7 ft/hr)	0.2–0.25 m (8–10 in)	3.4 MPa (500 psia)	34.4 m^3/min (1200 scfm)
Browning	Barre, VT	130 m (430 ft)	7.6 m/hr (25 ft/hr)	0.35–0.4 m (14–16 in)	0.86 MPa (125 psia)	34.4 m^3/min (1200 scfm)
Los Alamos	Pedernal, NM	30 m (110 ft)	6–7 m/hr (22 ft/hr)	0.35–0.45 m (14–18 in)	0.76 MPa (110 psia)	31.5 m^3/min (1100 scfm)

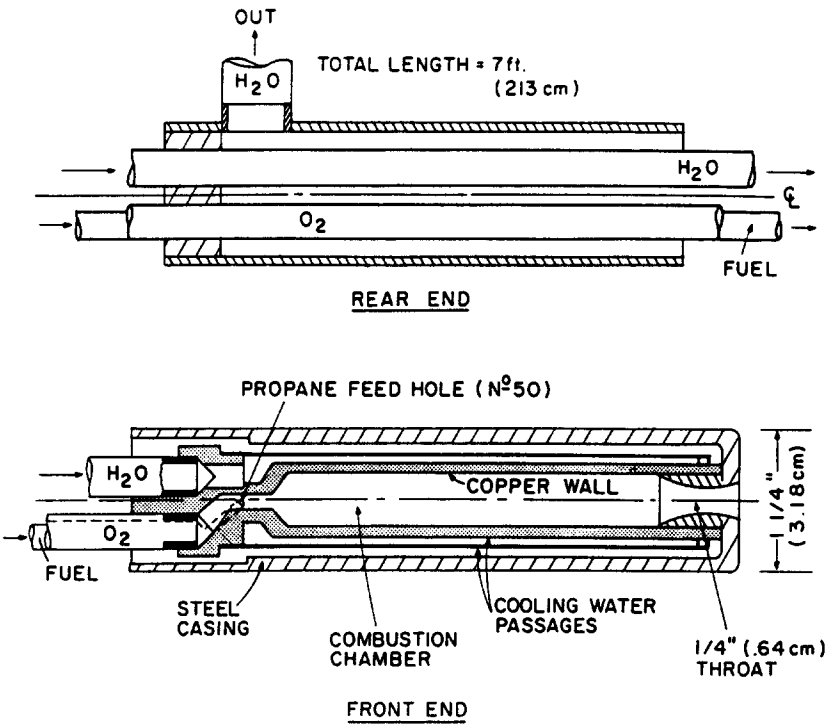


Fig. 3. Internals of burner used in Los Alamos experiments, shown schematically, with spark plug included for downhole ignition [1].

attempted many times. These efforts can be naturally divided into two categories: theoretical and quantitative estimations of the thermal stress required to spall a rock, and qualitative approaches based on examination of rock composition and petrography.

The majority of the early researchers focused on what was termed "spalling" but can actually be ascribed to ceramic fractures extending over much larger fractions of volume of the brittle material. In more recent years, after spallation became widely used for blasthole formation in surface mining and cutting in granite quarries, spallation was redefined to denote generation of thin chips, often smaller than a single grain, liberated from a rock body perhaps three orders of magnitude larger in scale as a result of much larger heat fluxes (0.5–10 MW/m^2) over relatively short times scales (0.1–1 sec). Not surprisingly, the number of alternative theories proposed and some confusion over what physical process was being examined has led to a good deal of misleading and contradictory papers about spallation. In this research, we focus on the present-day usual meaning of spallation, that is, formation of small rock fragments under high heat flux.

Norton [9] studied failure of brittle materials subjected to thermal shock and used the word "spalling" to describe rupturing of corners and ends of clay firebrick. As part of his investigation, spheres and bricks of a clay material were prepared, heated and cooled suddenly. Norton proposed that shear failure of the material was favoured under these circumstances, but his conclusions were formed from an incorrect physical basis [10]. Preston [11], a contemporary of Norton's, correctly pointed out that surface spalling must be a result of stresses acting on flaws parallel to the surface, generating much smaller fractures over a shorter time scale than those Norton had observed. Later, Preston [12] was able to produce thin, lamellar spalls in fireclay spheres through very rapid heating. As shown in the simplified sequence of events in Fig. 4, Preston's postulated mechanism relies on buckling of a thin slab of rock from the surface after the flaw has been extended by the compressive stress field. This seems to be the most reasonable explanation for spalling, though several alternative theories have been proposed since Preston's work 50 yr ago.

As Preston demonstrated, spalling relies on two distinct criteria concerning the heated sample dimension and its confinement: (1) in an unconfined specimen, the heated area must be a relatively small fraction (say, less than 10%) of the total area on that surface; thus, if no externally introduced lateral confinement is present, the far-field displacements at the surface must approach zero, thereby confining the heated portion; and (2) the heating rate must be large enough to force the surface temperature of the brittle material to attain high values before an appreciable fraction (again, say, less than 10%) of the volume is affected by heating; otherwise, unless the sample is confined against movement in all directions, stresses will be relieved through thermal expansion in the direction of least resistance. These fairly qualitative guidelines are highly situation-dependent and material-specific, but, in general, confinement of some fashion and a relatively large body of material are required for spallation to be favoured.

Spallation research was reinitiated in the 1960s by several parties including Gray [13] and Geller [18] of

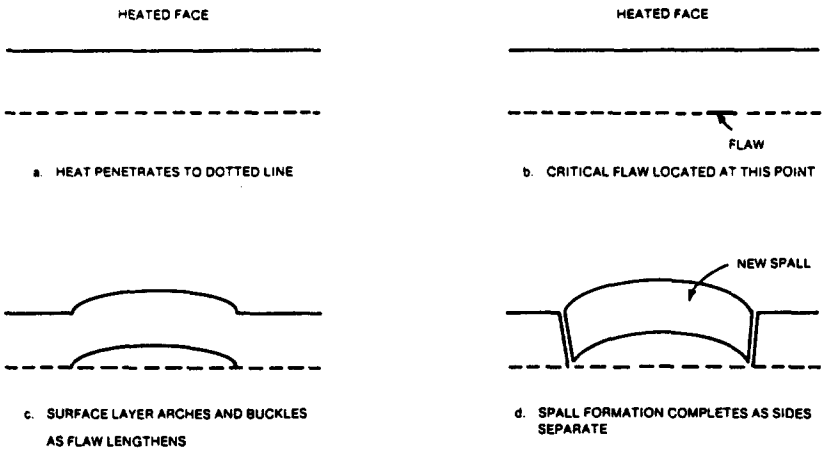


Fig. 4. Simplified chain of events leading to spall formation on surface of a semi-infinite solid [12].

the Canadian Mines Bureau. At that point, more quantitative arguments were sought to explain rock spallation by introducing the concept of a critical thermal stress or temperature rise required for spallation. Gray [13], of the Canada Mines Bureau, compared the plane compressive stress in a semi-infinite, homogeneous, elastic body confined against movement in the two directions parallel to the heated free surface to the compressive strength of the material (σ_c), deriving a rock spalling temperature change, ΔT_s, as:

$$\Delta T_s = \frac{(1-\nu)\sigma_c}{\beta_t E}, \tag{1}$$

in which E is Young's Modulus, ν is Poisson's ratio and β_t denotes the linear thermal expansion coefficient of the rock. For the 13 samples tested by the Mines Bureau, property values measured during the same investigation indicated that ΔT_s is essentially constant. In Table 2 [14–17], a range of ΔT_s for several rock categories is listed and clearly some variation in spallation temperature is predicted. Within the broad classifications of rock tabulated, wide variations of thermoelastic properties are likely, and only a rough estimate of the surface spallation temperature should be extracted from this table. A more rigorous approach would require direct measurements of ν, β_t, E and σ_0 for a specific rock (say, limestone) and subsequent application of equation (1) to assess the relative spallability of *that* specimen, and not the entire subset of rocks classified under such broad headings as "limestone", for example.

Although Gray [13] and Geller [18] assumed that the edge of the spall is formed first with the base following, spallation typically occurs over a depth of, at most, a few grains, and rock rarely exhibits shear failure on that scale. Recent evidence suggests that shear breakage in much larger laboratory samples under triaxial compression is most likely a product of coalescence of abundant pre-existing cracks that have lengthened axially and not extended by sliding [19–22].

In another set of studies, the Russians [23,24] have empirically correlated their theoretical results to match field and laboratory data, but some mathematical development of spallation mechanics from thermal stress calculations has been attempted. Dmitriyev *et al.* [23] adopted a slightly more elaborate approach than Gray [13]. They supported the mechanistic conclusions of Preston nearly 40 yr earlier that relied on surface heating, compressive stressing parallel to the plane of the surface, and then buckling of thin plates, or lamellae, from the heated surface. They ignored the details of that process and focused on an accurate estimation of the near-surface compression.

By linearizing the quasi-1-D temperature and stress fields imparted by a convective heat source imposed within a circular region on the surface of a half-space, Dmitriyev *et al.* produced analytical expressions for spall thickness and potentially achievable drilling velocity. The spall thickness δ_s must support the entire linearized temperature difference ΔT_s:

$$\delta_s = k_r \Delta T_s / h(T_f - T_s), \tag{2}$$

where h is the flame-jet convective heat transfer coefficient averaged over the heated area, k_r is the rock thermal conductivity, T_f is the flame temperature, and T_s denotes the spallation temperature at the rock surface. From solution of the convective heating problem for short times, the dimensionless time to cause spallation $\chi\,(\equiv h^2\alpha_r t_s/k_r^2)$ was determined from

$$1 - \exp(\chi)\,\mathrm{erfc}\sqrt{\chi} = \frac{\Delta T_s}{T_f - T_0}, \tag{3}$$

where α_r is the rock thermal diffusivity and

$$\mathrm{erfc}(\chi) = \int_{\chi}^{\infty} e^{-u^2}\,du.$$

The drilling speed V_{dr} is merely the quotient of thickness and time, and defining $T' = [\Delta T_s/(T_f - T_0)]^{-1}$:

$$\frac{\delta_s}{t_s} = V_{dr} = \frac{h\alpha_r}{(T' - 1)\,k_r \chi},$$

χ (as a function of T') must be obtained from the transcendental equation for surface temperature [equation (3)] and ΔT_s is given by equation (1).

Even more detailed analyses of the stress and temperature fields below a nominally circular heated surface were reported by Kolodko [24] in Russia and Lauriello [25] and Nelson [26] in the U.S. Their efforts were focused on solving the coupled stress and temperature fields under the heated spot either analytically [25] or by finite element techniques [24,26]. Despite the rigor of their calculations, these approaches are misleading because the locations of maxima in the stress fields scale directly with the heated spot size. For example, Lauriello [25] and Nelson [26] attributed spallation to tensile stresses accumulating at a depth of about 0.8 spot diameters below the surface, thereby predicting very thick spalls, not the thin flakes normally seen in practice. Kolodko [24] used shear failure arguments similar to those of Geller [18]. He also proposed a criterion to determine whether or not a rock will spall based on arguments that, during heating, the rock should not melt or fail in tension but reach the shear failure limit instead.

Attempts have also been made to correlate rock compositions and petrographic structural physical properties to spallation behaviour. Soles and Geller [27], coworkers of Gray at the Mines Bureau of Canada, examined the effect of rock physical properties and composition on susceptibility to jet piercing and concluded, as had several others, that no single rock property can predict or determine spallability. Rock composition and structure are extremely important, but their influences are not easily quantified. However, because no petrographic evidence for the α–β transition in quartz at 573°C or the decomposition of dolomite was ever found, those transformations, once thought to be important in causing spallation [28], are certainly of little or no consequence.

Rock fragments examined by Soles and Geller ranged from dust to 3-cm dia flakes, but almost all were multigranular, indicating that the cleavage plane can often sever minerals along intragranular weaknesses and that spallation is controlled by the thermomechanical behaviour of all minerals comprising the rock. Quartz and nepheline promote spallation, while mafic, micaceous and other soft, pliable minerals have an adverse effect on spallability [29]. In Geller's later compendium of spalling work to that date [18], he hypothesized that the role of these mineralogical and structure-related factors were not as important as their effects on the thermophysical properties of the rock specimen, thereby supporting a property-based criterion.

In spite of plentiful attempts to formulate a comprehensive spallation mechanism, no satisfactory quantitative description of spallation micromechanics that agrees with experimental evidence has been proposed. All previous theories incorporate an unrealistically simple description of the relevant rock failure mechanisms. Certainly, examination of rock microstructure is crucial for determination of spall characteristics, but a detailed examination of grain–grain interactions is unreasonable and too complicated. Analytical methods incorporate only macroscopic thermophysical properties, such as uniaxial compressive strength and Young's

Table 2. Average rock properties and spalling ratings of some common rock types

Rock	Rating	$\beta_t(\times 10^{-6})(K^{-1})$	E(GPa)	$(\rho C_p)_t(\times 10^6)\left(\frac{J}{m^3-K}\right)$	ν	σ_c(MPa)	$\frac{\sigma_c(1-\nu)}{\beta_t E}$	$\frac{\beta_t E}{(\rho C_p)_t \sigma_c}\left(\frac{m^3}{J}\right)$
Basalt	Fair–Poor	4–5	30–50	2.5–3.3	0.22	80–300 (130)	560	4.6×10^{-10}
Diorite	Good–Poor[1]	6–7	35–70 (60)	3.4	0.25	100–200 (120)	230	9.6×10^{-10}
Dolomite	Excellent	11–13	50–100 (70)	5.2	0.20	50–150 (100)	95	16.1×10^{-10}
Gneiss	Good–Fair[1]	5–9	20–70 (40)	2.5	0.22	150–200 (180)	500	6.2×10^{-10}
Granite	Good–Fair[1]	8–14	20–70 (50)	2.5	0.20	100–250 (200)	320	10×10^{-10}
Limestone	Poor	8–11	10–70 (30)	4.0	0.20	10–200 (80)	230–700[2]	$4–9 \times 10^{-10}$
Marble	Fair	7	30–80 (40)	2.5–2.7	0.30	60–200 (100)	250	10×10^{-10}
Quartzite	Excellent	11–15	20–100 (50)	2.7	0.25	140–300 (200)	230	12×10^{-10}
Sandstone	Good–Poor[1]	10–13	7–40 (20)[3]	2.5	0.27	30–250 (180)	550	5×10^{-10}
Schist	Fair–Poor	6(?)	40–90 (50)	2.7	0.15	10–200 (100)	140–280	$10–20 \times 10^{-10}$
Tuff	Poor	3–5	1–10 (7)	2.7	0.10	10–30 (20)	640	5.2×10^{-10}

[1]Depending on quartz content.
[2]Despite this result, no spallable limestone has ever been found.
[3]Not including quartzitic varieties.
Numbers in parenthesis are typical average or representative values. References [14–17].

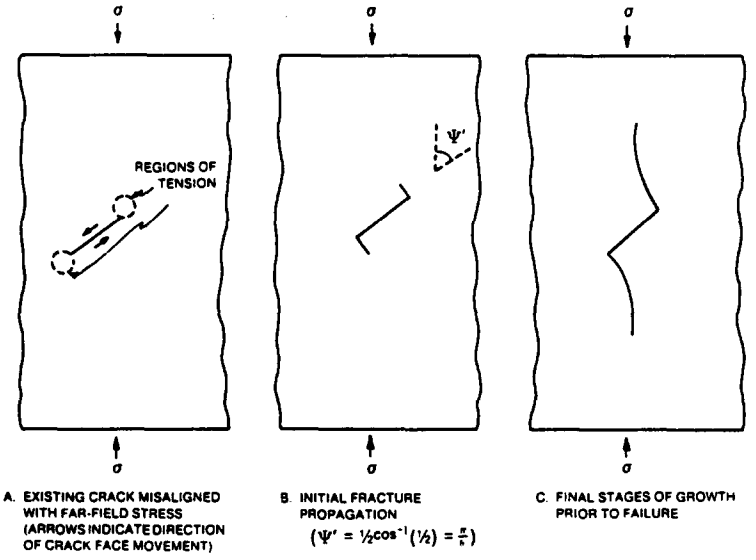


Fig. 5. Nonplanar crack extension induced by far-field applied compressive stress [20].

modulus, measured typically without regard to the spatial distribution of applied stress or character of grain–grain thermal interactions. On the other hand, petrographic or compositional analyses are useful only for establishing qualitative guidelines for determining whether or not a rock is spallable, but lend no insight into quantitative descriptions of rock spalling characteristics, such as surface temperature and average spall thickness.

ROCK FAILURE MECHANISMS APPLICABLE TO SPALLATION

To begin our analysis of the problem, we consider the fairly simple stress and temperature fields for the configuration relevant to spallation. If the heated rock body can be considered to be an infinite half-space ($z > 0$ for all x, y) with a 1-D temperature wave travelling into it, the compressive stresses (σ_{xx}, σ_{yy}) parallel to the surface are merely proportional to the temperature rise at that depth [30, 31]:

$$\sigma_{xx} = \sigma_{yy} = \frac{\beta_r E \Delta T}{1 - \nu}, \tag{4}$$

from which equation (1) can be derived [13].

Determination of rock stresses in no way demonstrates at what stress rock will fail or fracture. Thus, without a suitable theory of rock fracturing under stress, prediction of spallation behaviour is impossible. Historically, researchers have assumed that flaws inherent in the material were responsible for microscopic breakdown and eventual macroscopic disintegration of a laboratory sample. Griffith [32] was one of the first to recognize that consideration of pre-existing rock flaws was extremely important in formulating and evaluating failure criteria.

Our estimation of the stresses responsible for spallation revealed that they were entirely compressive [equation (4)]. Griffith showed that a single crack aligned at an arbitrary angle to the compressive stress will extend due to tension at the tip of the crack and, as Nemat-Nasser and Horii [20] calculated analytically, attempt to orient with the direction of that compressive load (Fig. 5). This process of extension was assumed by Griffith to be unstable, but in the absence of any lateral tension across the crack face as it lengthens, growth is stable until the final stages of finite sample loading. In practice, under compressive load, the presence of many interacting microcracks all extending in the same direction will potentially initiate buckling of the specimen parallel to the nearest free surface, resulting in exfoliation or rockburst of the material from the main body [33, 34].

Simple, global failure models such as that proposed by Griffith [32] and modified by McClintock and Walsh [35] fail to recognize that a distribution of cracks exists in a rock specimen. These theories merely predict failure when the applied stress reaches a critical value ($\sigma = \sigma_c$). If a single or a few flaws are responsible for initiating failure, it is not reasonable to assign a unique value of the compressive strength σ_c to the rock for all loadings. In spallation drilling, the highest stress as predicted by equation (4) occurs at the surface, giving the nonsensical result of zero thickness spalls.

Thus, for analysis of spallation mechanisms, the two major drawbacks of these fracturing criteria are: (1) no provision is made for flawed microstructure and granular fabric of the material (i.e. it is assumed that flaws exist, but their number, size and shape distributions are not considered, and thus the shape of the specimen and the manner of loading are inconsequential); and (2) repeated experimental tests yield a distribution of failure strengths, rather than a single value. Consequently, Weibull's [36] statistical failure theory is usually more appropriate for describing thermal shock experiments [37].

Because extension and coalescence of microcracks are responsible for rock failure, determination of the size

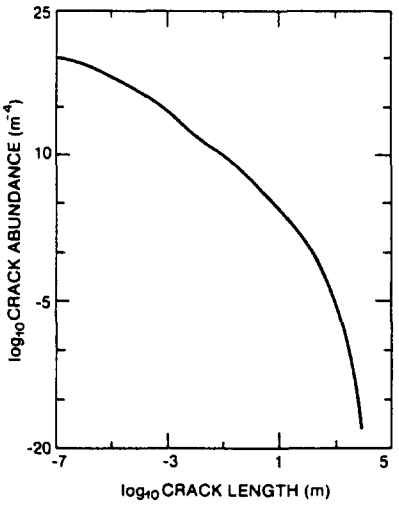


Fig. 6. Distribution of measured crack sizes as a function of length [40].

and number of inherent flaws in a rock sample is important. Consistent with a statistical approach, the strength of a sample has been experimentally correlated with the size of the specimen. This can be directly related to the number of large microcracks present. Because a smaller sample will contain, on average, fewer critical-sized flaws, the stress required to extend these flaws grows as laboratory sample size decreases. Using Hadley's [38] scanning electron microscope measurements and Peng and Johnson's [39] optical microscope analysis, Dey and Halleck [40] augmented their own observations of geologic outcrops to estimate the flaw density within intact cores and field specimens (Fig. 6). Their measurements indicate that cracks less than 1 m in size follow a numerical density distribution of:

$$\Psi(l) = Al^{-3}, \tag{5}$$

where A is a constant, l is the crack length and $\Psi(l)$ is the cumulative number of cracks of length l or less. As Dey and Halleck [40] show, this exponent (-3) corresponds to a critical value in the microcrack interaction strength. Thus, for sample sizes up to 1 m, microcrack interactions appear to be responsible for rock failures. At larger sample sizes, the inclusion of progressively larger cracks does not appear to affect the failure characteristics. Size-dependent strength can be envisioned, therefore, as a consequence of inherent microcrack distribution.

A complicating factor for analysis of thermal spallation is the possibility of creating more microcracks than were initially present at ambient temperature. The multimineralic nature of rock with irregular or angular grain boundaries leads to extremely complex stress behaviour on a microscopic scale, often causing sufficiently concentrated tension to initiate intergranular cracking. In the past, prediction of increase in crack density with heating has been required for studies of thermal expansion behaviour in rocks and porosity changes with temperature [41,42]. The stresses due to grain–grain interactions are separate from those induced by overall temperature changes in the body and arise primarily from mismatched coefficients of thermal expansion ($\beta_1 \neq \beta_2$) between neighbouring grains. Simplified models of actual mineral geometry have been adopted to estimate the importance of stresses produced by this mechanism in instances where a totally deterministic approach would be prohibitively complicated [43,44]. In all of these simple models, thermal expansion coefficients and Young's modulus of the constitutive minerals dictate, to first-order, the level of interactions among grains and pre-existing cracks.

Assumed geometries include two grains of different properties cemented along a common boundary and one grain included in a continuous matrix of another mineral. In the latter case, if the confining pressure exceeds the crack-closing pressure, then dilatancy from grain expansion cannot occur until this external pressure is overcome by stresses attempting to form new cracks. If the bonded joint between the grain and the matrix has any residual tensile strength, then this must also be surpassed. Edwards [43], Eshelby [44] and Johnson *et al.* [41] showed that the magnitude of these stresses is about:

$$\sigma \simeq E\Delta\beta\Delta T, \tag{6}$$

regardless of geometry, where $\Delta\beta$, the difference in the thermal expansion coefficient between grains, can be about $5 \times 10^{-6}\,\mathrm{K}^{-1}$, and ΔT is the overall temperature change induced in the sample, not the difference between matrix and inclusion temperatures. Typically, the stress needs to be several hundred MPa to match the confining capability of the surrounding medium if all cracks are initially closed. However, near a free surface, a 40°C temperature rise can induce tension around the inclusion (if its β is larger) of about 10 MPa, large enough to initiate secondary crack formation and weaken the sample [42]. The resulting crack density, greater than that at room temperature, will enhance rock's spallability and permit failure at a lower stress than calculated by using only room temperature properties.

The interaction of an included grain and an existing microcrack can also be extremely important. The problem of one crack of finite length and one inhomogeneity has been solved by Atkinson [45], who approximated the change in stress intensity factor ΔK_I induced as a result of heating as:

$$\Delta K_I \simeq 9.45\,\mu\Delta\beta\Delta T\sqrt{l'}, \tag{7}$$

where l' is the initial crack length, and μ is the shear modulus. K_{Ic}, the critical stress intensity factor, can be expressed as:

$$K_{Ic} = \frac{4\gamma\mu}{1-\nu}. \tag{8}$$

Determination of γ, the fracture specific energy, is critical to use of this formula, but whether the individual grain's value or that for the whole rock is used remains uncertain. If we assume values of $\gamma = 1\ \mathrm{J/m^2}$ [42], $\mu = 30\ \mathrm{GPa}$, $\Delta\beta = 5 \times 10^{-6}/\mathrm{K}$ and $\Delta T = 40°\mathrm{C}$, equation (8) indicates that cracks longer than about 40 μm will be strongly affected by the presence of an inclusion because $\Delta K_I \geqslant K_{Ic}$. More study is required to

generalize this effect for more complex and realistic geometries for polycrystalline granular materials.

In theory, use of these analytic equations in conjuction with the propagation speed and flaw size distribution permits determination of cracking behaviour of a flaw in an otherwise homogeneous medium. Unfortunately, such a totally deterministic approach is not tractable because: (1) the medium surrounding the flaw is not homogeneous and could impart large deviations in the local thermal gradient; (2) propagation is not proceeding predictably through the medium; and (3) fracture specific energy is highly dependent on the exact location of the flaw and the presence of other heterogeneities in its vicinity [42].

Numerical techniques are required at this point to examine thermomechanical effects in greater detail and assess the importance of other parameters that may influence rock spalling. Effects of nonuniform material properties, widely variant boundary conditions of both temperature and stress, and irregular and complex geometries were examined with Browning and Anderson's [46] finite-element code TSAAS (thermal and stress analysis of plane and axisymmetric solids).

Because element-by-element changes in material properties and material configuration are permitted, we were able to consider grain–grain interactions. Although sharp discontinuities across grain boundaries are not physically realistic, the results of perfect intermineral cementing represent an upper bound on the accumulated local stress. As an example, consider the numerically computed stress distribution surrounding an idealized quartz grain that has a higher coefficient of thermal expansion than the surrounding space of feldspar (Fig. 7). If the surface of this semi-infinite confined solid is heated, the quartz will attempt to expand axially and thus pull the feldspar apart in tension. Exactly where or at what angle a fracture would start is difficult to specify, but conditions for crack initiation are present if a weakness in the fabric is encountered in that region of high tension (σ_t for granite $\approx$ 10 MPa).

Although many spalls actually observed in the field contain more than one grain, the complete collection of spalls from a laboratory spallation test conducted by the Canadian Bureau of Mines [18] revealed that the average spall thickness of 0.4 mm was less than a representative grain size. Thus, individual grains are split quite regularly during spallation and contribute significantly to overall chip production. A detailed analysis of the mechanics of a polygranular system is highly sample-dependent, but the estimates shown indicate that this mechanism could be important.

Another manner in which cracks can initiate involves perturbation of the stress field by irregularities in the rock surface profile. Surface roughness, if very pronounced, can induce tensile stress acting on the plane parallel to the mean surface, which can be computed on the assumption of a homogeneous, isotropic, elastic media [10]. Our examination of the surface of a spalled granite sample revealed that the associated roughness is, as one expects, of the order of a grain size in amplitude

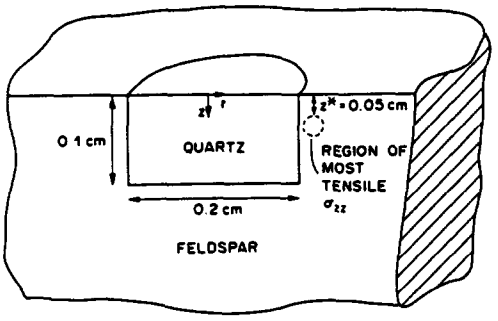


Fig. 7. Model of an included, cylindrical quartz grain, showing region of greatest tension within surrounding feldspar, induced by mismatched thermal expansion coefficients.

and one or several grains in wavelength, meaning that the stresses calculated in this manner are acting on the same scale as those induced by the effects of multigranular structure. The tension resulting from the combination of these processes was computed with TSAAS to be sufficient to form new cracks.

In summary, multimineralic and roughness effects are substantial enough to initiate microcracking on a granular level. Eventual extension of these new flaws and pre-existing defects by a predominantly compressive stress field is ultimately responsible for spallation. At this point, these crack, initiation mechanisms need to be incorporated into a more phenomenological theory to predict spallation stresses and resulting spall sizes.

APPLICATION OF WEIBULL'S STATISTICAL THEORY OF FAILURE TO SPALLATION

Weibull [36] proposed a statistical failure theory in an attempt to account for the dependence of failure strength on sample size. Because materials inherently contain flaws, a sample can never achieve its true potential strength. Thus, if a region is stressed, a "risk of rupturing" is incurred that depends on flaw distribution, the size of the stressed zone and the stress level within that domain. A larger flaw cannot accept as much stress before it starts to fail, but the probability of encountering such a flaw in a small region is less than the chance of locating a critical defect within a large sample. Therefore, Weibull's criterion reduces to a distribution of strengths dependent on the stressed volume.

Tensile failure of brittle materials, which is based on the assumption that only the most critical flaw induces macroscopic fracturing, is consistent with predictions from Weibull's theory [47]. Similarly, compressive failure near a rock surface during spallation should involve, at most, a few flaws, unlike macroscopic failure of a laboratory sample that is several cm in diameter and relies on the interaction of many flaws. Equivalent compressive failure criteria have been proposed and experimentally tested for coal crushing [48] and granite breakage [49]. In fact, most of the reported size–strength relationships for compressive failure are merely restate-

ments of Weibull theory. Thus, use of Weibull statistics seems somewhat justified for modelling spallation, especially under high heat fluxes where dimensions of flaws approach the heated thickness.

The Weibull distrubiton of size-dependent strengths can be represented as a probability distribution [47]:

$$G(\sigma) = 1 - \exp\left[-\int_0^V \left(\frac{\sigma}{\sigma_0}\right)^m \mathrm{d}V \right] \equiv 1 - \exp(-I). \quad (9)$$

At median spalling conditions, G, the cumulative failure probability is 0.5 ($I = 0.693$), which will be used for the sake of argument in the following discussion [50]. With a finite m, a failure probability of 1 results under infinite stress if the material contains any flaws. The Weibull parameters, m and σ_0, are obtained from laboratory experiments that examine failure stress for several experiments with the same sample configuration. A higher value of m indicates a more spatially uniform rock. For a perfectly homogeneous material $m \to \infty$ and all induced failures occur when $\sigma = \sigma_0$, the applied load. σ_0 can be thought of as the intrinsic strength of a sample of 1 unit dimension in volume, or 1 m^3 in SI units, for example.

To determine the stress distribution properly, the temperature field must be specified. Because spallation is a periodic process removing mass and energy, the stress and thermal field calculations are coupled through determination of spallation thickness and a complex iterative procedure is required. Such calculations will rigorously produce successive spall dimensions and temperature characteristics, but the problem can be reduced to a quasi-steady-state temperature and stress analysis with a moving boundary similar to a continuous sublimation process [10, 13]. If the discrete jump in temperature caused by chip liberation (calculated to be less than 30% of overall ΔT [10]) is ignored, it can be assumed that the rock merely disappears at a velocity u_r in the direction normal (x) to the mean surface, and

$$k_r \frac{\partial^2 T}{\partial x^2} = -u_r \frac{\partial T}{\partial x} (\rho C_p)_r. \quad (10)$$

The solution for a specified surface temperature rise of spallation ($\Delta T_s \equiv (T_s - T_{r0})$ is:

$$T = (T_s - T_{r0}) \exp(-u_r x/\alpha_r) + T_{r0}, \quad (11)$$

where α_r is the rock's thermal diffusivity. The assumption of constant surface temperature rise ΔT_s is based on quasi-steady-state conditions in which the rock reaches a specified surface temperature and stress and then spalls. ΔT_s is a function, albeit a weak one, of applied heat flux. The thermal penetration distance α_r/u_r is of the order of 1 mm for reasonable penetration rates [0.0017 m/sec (20 ft/hr)], justifying the plane heating assumption in the thin heated skin.

The integral in equation (9) must be performed over the volume that is affected by heating and that will potentially remove a certain point on the rock surface. Immediately below that point, the temperature field is given by equation (11). The region that must contain the critical flaw is a cone with its apex at the surface and a radius of $C_L x/2$, where x is the normal distance into the rock and C_L is the diameter of an average chip divided by the thickness, i.e. the aspect ratio. Thus, the differential volume is $\pi(C_L x/2)^2 \mathrm{d}x$ and, at median spalling conditions [$G(\sigma) = 0.5$]:

$$0.693 = \left(\frac{\sigma_s}{\sigma_0}\right)^m \frac{\pi C_L^2}{4} \int_0^\infty x^2 \exp(-\omega x)\, \mathrm{d}x, \quad (12)$$

where $\omega = m u_r/\alpha_r$. Furthermore, $Q_r = -k_r(\mathrm{d}T/\mathrm{d}x)_{x=0} = (\rho C_p)_r u_r (T_s - T_{r0})$ [from equation (11)] and

$$\sigma_s = \frac{\beta_r E \Delta T_s}{1 - \nu},$$

from equation (4). These two expressions can be combined with equation (12) to produce the following expression for the applied heat flux Q_r:

$$Q_r = (\rho C_p)_r \left(\frac{1-\nu}{\beta_r E}\right) \sigma_0 \left(\frac{2(0.693)}{C_L^2 \pi}\right)^{1/m} \left(\frac{m u_r}{\alpha_r}\right)^{3/m} u_r. \quad (13)$$

Finally, the rock surface temperature at any point can be calculated by combining the equation $Q_r = (\rho C_p)_r u_r (T_s - T_{r0})$ and equation (13) to yield:

$$T_s - T_{r0} = \left[\left(\frac{Q}{\rho C_p}\right)_r^3 \left(\frac{(1-\nu)\sigma_0}{\beta_r E}\right)^m \times \left(\frac{2(0.693)}{\pi C_L^2}\right) \left(\frac{m}{\alpha_r}\right)^3 \right]^{1/(m+3)}. \quad (14)$$

An aspect ratio must be assumed, and a value between 8 and 15 for C_L seems representative of the chips Geller collected [18].

Buckling theory provides another method of determining the diameter-to-thickness ratio [50]. An incipient chip formed by subsurface cracking will spall by a buckling mode of failure [12, 30] from the surface that is confined against movement in the plane of a crack extension when the cracked length d divided by the thickness of the potential spalls exceeds

$$\frac{d}{\delta} = \pi \left[\frac{E}{6(1-\nu^2)\sigma} \right], \quad (15)$$

which gives:

$$C_L = \frac{d}{\delta} = \frac{\pi}{6(1+\nu)\beta_r \Delta T_s} \simeq 10. \quad (16)$$

Thus, $C_L = 10$ will be assumed for the remainder of this development.

From equation (14), high heat fluxes induce stresses in a thinner skin of rock more quickly. The overall stress level in that region must be higher, increasing the probability that the shorter flaws encountered in this layer will extend to a critical size. For reasonable values of Q_r(1×10^6 to 1×10^7 W/m^2) and rock properties [$(\rho C_p)_r = 2.64 \times 10^6$ J/m^3 K, $E = 45$ GPa, $\nu = 0.25$, $\beta_r = 10^{-5}$ K^{-1}, $\sigma_0 = 70$ MPa, $m = 20$, $\alpha_r = 10^{-6}$ m^2/sec), ΔT_s is predicted to be in the range of 400–550°C.

Although Weibull theory predicts no strength at infinite sample size, its use is proposed here for compressive failure over the size range relevant to spallation,

namely, 0.1–1 mm, and not for typical laboratory specimens where many flaws are active in producing rock failure. Difficulty arises in determination of the Weibull parameters, m and σ_0, which are typically obtained by mechanical, not thermal, loading of laboratory rock samples. Because the process of heating rock can induce more cracking, especially near the surface, than was present initially, flaw structure and distribution in the rock will be affected. Therefore, both m and σ_0 will be functions of temperature and of position as well, because heterogeneity and strength are rapidly varying within the rock. Weibull parameters should be measured as functions of the mean rock temperature; but if the full temperature difference occurs over the distance of one or two grains, those values will need to be measured by causing failure on that scale. Methods of measuring temperatures at incipient spallation will be discussed in the next section of the paper, and the vailidity of Weibull's theory for this application will be assessed by comparing predicted and actual surface spallation temperatures.

Given the simplification presented in the previous section developed in collaboration with Dey [50], the distribution of spall sizes expected during any drilling operation can now be predicted as a function of drilling rate, rock properties and Weibull parameters. The probability that a spall is of a thickness less than or equal to x, $G_1(x)$ can be expressed in terms of the distribution of Weibull strengths, $G(\sigma)$. At median spalling conditions,

$$G(\sigma) = 1 - \exp(-I_o)$$
$$= 1 - \exp\left(-\int_V (\sigma/\sigma_0)^m \, dV\right) = 0.5. \quad (17)$$

$G(\sigma)$ can also easily be restated in terms of a spatial failure distribution $G(x)$, because σ and distance into the rocks are uniquely related by equations (4) and (11). From the integration of I from zero spall thickness to x and using $I = 0.693$ [i.e. cumulative probability $G(\sigma) = 0.5$] at median spalling conditions, if $x \to \infty$:

$$\frac{\pi C_L^2}{2}\left(\frac{\sigma_s}{\sigma_0}\right)^m \left(\frac{\alpha_r}{m u_r}\right)^3 = 0.693. \quad (18)$$

$G_1(x)$, the normalized distribution of chip sizes, is just $G/0.5$ at median spalling conditions, or:

$$G_1 = 2 - 2\exp\{-0.693[1 - (1 + \omega_1 + \omega_1^2/2) \times \exp(-\omega_1)]\}, \quad (19)$$

where $\omega_1 = m u_r x/\alpha_r$ and u_r is the normal penetration rate [50]. Notice as $\omega_1 \to \infty$, G_1 approaches 1, giving the proper normalization. Extending Dey's analysis by differentiating equation (19), the chance that a spall will be formed with thickness between ω_1 and $\omega_1 + d\omega_1$ [$\equiv P(\omega_1)$] is:

$$P(\omega_1) = 0.693\,\omega_1^2 \exp[-\omega_1]\exp\{-0.693 \times [1 - (1 + \omega_1 + \omega_1^2/2)\exp(-\omega_1)]\}. \quad (20)$$

The volume flux at any differential area on the hole surface is merely the product of normal penetration rate (u_r) and the area that is spalling. Therefore, if N_i is the number of chips per unit time and unit area of spalled surface and $V(x)$ the volume of a chip is $\pi C_L^2 x^3/4$, then

$$N_i = \frac{4m^3 u_{ri}^4}{\alpha_r^3 \pi C_L^2 \int_0^\infty \omega_1^3 P(\omega_1)\, d\omega_1}. \quad (21)$$

The distribution of chips leaving the hole per unit time within a unit dimensionless size interval is defined as Q and can be evaluated numerically [10] from:

$$Q = \frac{\int_A N_i P(\omega_1)_i \, dA}{N_{tot}}, \quad (22)$$

where the distribution has been normalized with the total number of chips liberated, N_{tot}.

To proceed further requires an assumption about the shape of the hole being drilled to integrate over its area. For the sake of this derivation, a hemisphere of aribtrary size advancing at a rate of V_{dr} was chosen. Therefore:

$$u_r = V_{dr}\cos\theta, \quad \omega_1 = m V_{dr}(\cos\theta)x/\alpha_r = \xi_0 \cos\theta,$$

$$N_i\, dA_i = \frac{4m^3 V_{dr}^4 \cos^4\theta_i (2\pi R_h^2 \sin\theta_i\, d\theta_i)}{\pi \alpha_r^3 C_L^2 I_3},$$

$$N_{tot} = \frac{8m^3 V_{dr}^4 R_h^2}{5 C_L^2 \alpha_r^3 I_3}, \quad (23)$$

where θ is the angle that the normal unit vector forms with the vertical at any position and

$$I_3 = \int_0^\infty \omega_1^3 P(\omega_1)\, d\omega_1.$$

$Q(\xi_0)$ was then integrated using the trapezoid rule to evaluate the median, mode, number mean (n.m.) and weight mean (w.m.) of dimensionless spall thicknesses for $C_L = 10$, which are indicated on Fig. 8 and defined as:

$$|\xi| = \xi_{0(med)} = \xi_0 \text{ such that } \int_0^{\xi_0} Q(u)\, du = 0.5 = 2.9,$$

$$\xi_0^{\circ} = \xi_{0(mode)} = \xi_0 \text{ at a maximum value of } Q(\xi_0) = 2.0,$$

$$\langle \xi_0 \rangle = \xi_{0(n.m.)} = \int_0^\infty u Q(u)\, du = 3.35,$$

$$\langle \xi_0^3 \rangle^{1/3} = \xi_{0(w.m.)} = \left[\int_0^\infty u^3 Q(u)\, du\right]^{1/3} = 4.85.$$

Note the direct dependence of the moments of distribution on m and the inverse dependence of spall thickness on drilling rate ($x = \xi_0 \alpha_r/m V_{dr}$) and, therefore, to first-order, on the rate of heat transfer. For some representative values of physical parameters and drilling rate ($V_{dr} = 0.0017$ m/sec, $m = 20$, $\alpha_r = 1.2 \times 10^{-6}$ m^2 sec), these expressions yield the following numerical values for the distribution averages of the chip size:

$$x_{med} = 0.10 \text{ mm},$$

$$x_{mode} = 0.07 \text{ mm},$$
$$x_{n.m.} = 0.12 \text{ mm},$$
$$x_{w.m.} = 0.17 \text{ mm}. \quad (24)$$

If the reliability of the Weibull distribution for spallation fracture mechanics is verified in future investigations, careful collection and characterization of rock chips might prove useful as a quick method of estimating m. The analysis would be complicated by complex breakage of spalls that can often occur as they leap from the heated surface.

Both phenomenological Weibull cofficients, m and σ_0 are not directly obtainable from the chip size distribution. The range of spall sizes does serve as a check for these parameters that are determined merely from the distribution of strengths at which the samples fail. For example, from compressive failure testing of Berkeley blue granite samples, Weibull parameters of $m = 20$ and $\sigma_0 = 69$ MPa-m$^{3/20}$ were determined by Dey and Kranz [51]. With these values, it is possible to determine drilling rate, temperature difference and heat flux as functions of other rock properties and any one of the other two variables. In addition, u_r can be expressed as $V_{dr} \cos\theta$, in which V_{dr} is the drill penetration rate and θ is the angle that the hole surface makes with the horizontal.

The applicability of this analysis to a range of heat fluxes can be demonstrated by applying equation (14) to spalling in an underground diesel exhaust passage, initially approx. 2 m in diameter in gneissic rock [13]. Using the estimated penetration rate induced by the 300°C gases ($u_r \simeq 0.006$ m/hr), a spallation temperature of 150°C is computed [$(\rho C_p)_r = 2.64 \times 10^6$ J/m^3-K, $\beta_r = 10^{-5}$ K^{-1}, $E = 45$ GPa, $\alpha_r = 10^{-6}$ m^2/sec, $m = 20$, $\sigma_0 = 70$ MPa]. Considering the uncertainties involved in evaluating relevant properties and rate of rock removal, this agrees favourably with Gray's heat conduction analysis that crudely estimated a surface temperature of 110°C. Gray's computation was based on the observed spall thickness of 2.5 cm but also relied heavily on ascertaining the time of first spallation. Despite the reasonable agreement demonstrated here, Weibull's statistics should be applied with caution when thick slabs or spalls are generated or expected, because then many flaws must be active, violating one of the major assumptions of the theory.

EXPERIMENTAL TESTING OF WEIBULL THEORY APPLIED TO ROCK SPALLATION

Experiments have been conducted to test the reliability of our spallation mechanics theory. The surface temperature at incipient spallation has been measured indirectly as a function of applied heat flux and rock type for two varieties of granite, Westerly and Barre. Well-characterized sources of heating were supplied by both a CO_2 laser and a propane welding torch and directed at smooth surfaces. After estimating the actual heat flux delivered to the rock surface from these two heat

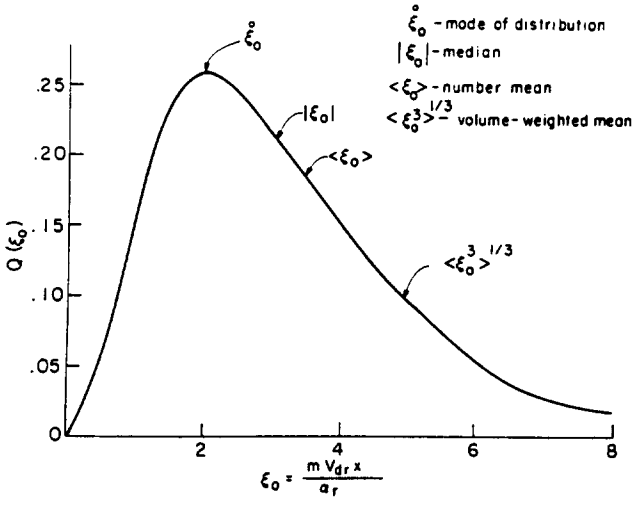


Fig. 8. Distribution of spall sizes exiting a hypothetical hemispherical hole, normalized with the total number of spalls generated per unit time.

sources, the surface temperatures present at spallation were calculated and compared to those expected from both the Weibull theory and previously reported spallation temperatures.

From solution of the transient heat conduction equation in one dimension for a constant heat flux on a semi-infinite solid:

$$T - T_{r0} = \frac{2Q}{k_r}\sqrt{\alpha_r t}\ \text{ierfc}\frac{x}{2\sqrt{\alpha_r t}} \quad (25)$$

and from Weibull statistics at a set spall diameter-to-thickness ratio C_L and median spalling conditions [$P = G(\sigma) = 0.5$] [cf. equation (14)], the surface temperature rise becomes:

$$\Delta T_s = \left[\frac{4(0.693)\,(Q/k_r)^3}{\left[\int_0^\infty u^2[\text{ierfc}\,u]^m\,du\right] \times \left[\frac{\sqrt{\pi}\,\beta_r E}{(1-\nu)\sigma_0}\right]^m \pi^{5/2} C_L^2}\right]^{1/m+3}, \quad (26)$$

where ierfc(u) is the integral complementary error function. Therefore, for a specified set of thermophysical properites (β_r, k_r, α_r, E, etc.), if the heat input to a rock (Q), m and σ_0 are known, the surface temperature at median conditions for spall formation is calculable. This computed value can then be compared to a measured surface temperature at the onset of spallation induced by rapidly heating granite. Once the time required to form the first spall and the heat flux transmitted to the block are determined, $\Delta T_s[T(x=0) - T_{r0}]$ can be calculated as $\Delta T_s = 2Q/k\sqrt{\alpha_r t/\pi}$. The dependency of predicted surface temperature on applied heat flux is often slight ($\Delta T_s \alpha (Q)^{3/m+3}$) but identical to that for rock drilling. In particular, for extremely homogeneous samples ($m \gg 1$) such as the granites used in these tests, variation with heating rate should be minimal.

The laser and propane torch heating sources used here each have experimental advantages and disadvantages, but they complement each other well. Heat flux from a high-energy continuous CO_2 gas laser is easily determined. Because previous work has indicated that nearly all of the i.r. energy from the CO_2 source is absorbed by the rock and very little is reflected [25,26], the total amount of power delivered per unit area is directly obtainable through a simple burn-pattern experiment. However, the portion of the beam diameter containing the majority of energy for most lasers, including the apparatus used in these tests (total diameter $\simeq 6$ mm), is often not much larger than the expected spall diameter. Therefore, unrealistically small spalls could be produced with laser heating. For flame-induced spallation, the chips produced and, therefore, the spalling temperature should be more representative of those from an actual drilling operation. The net available heat flux issuing from the torch nozzle may be readily calculated. However, the amount of heat actually delivered per unit area to point on the surface as a function of gas flow rates, distance from the torch nozzle to the flat surface, and radius along the rock surface from the centerline stagnation point where the gas first impinges must be determined from a separate experiment.

One of the simplest means of obtaining measurements under these severe conditions relies on the solution to the transient heat conduction equation [equation (25)]. In order to use this relation, the heat flux at the point of spallation and the time at spallation are needed. The latter quantity is easily obtained by recording the entire heating and subsequent spallation process on high-speed videotape (2000 frames/sec), and reviewing the recording. A separate calibration experiment using the torch to heat a highly-conductive, nonspallable material with well-known thermophysical properties must be performed to estimate the heat flux distribution.

PROPANE HEATING TESTS

Granites from Westerly, Rhode Island and Barre, Vermont, were chosen as standards from which to gather experimental data because they are two of the best-characterized and most frequently studied varieties. Diamond-saw cut blocks were subjected to a high-temperature jet of combusted propane and oxygen, impinging normal to the largest available working face. After two or three experiments had been conducted on a surface, the block was cut again approx. 4 cm below the damaged layer to expose fresh rock for further testing.

High-speed videotaping of the entire spallation experiment provided easy determination of the time required for rock to spall from a starting condition of ambient temperature. A Spin Physics 2000 video system was selected because it allows fairly high resolution monitoring of the events viewed by the high-speed video camera. A movie camera lens was focused on the granite block surface to be spalled, usually producing a field of view of about 7.5 cm square. The torch, an Airco propane heating burner (model 800) equipped with the largest capacity tip available (No. 757-18, 2.5 cm dia), was mounted in position above the rock surface such that the gas flow issued approximately perpendicular to the test area, and the standoff distance was either 15.2 cm (6 in) or 20.3 cm (8 in).

The flows delivered to the torch mixing chamber were controlled by commercially available gas cylinder regulators. Once the torch was lit with a welding sparker, oxygen flow was monitored with a rotameter and the propane delivered was regulated to maintain a flame tip length of approx. 2.5 cm, as suggested by the manufacturer for stoichiometric flow conditions. After the videotape speed had reached 2000 frames/sec, recording was started. Spallation commenced within 2 sec after exposure of the surface for all experiments conducted and was allowed to continue until a region about 8–10 cm in diameter had been damaged by spallation.

Although the momentum provided by the impinging jet was considerable, most spalls were ejected discretely and violently from the surface, releasing considerable strain energy and jumping a few feet upward. Thus, some qualitative evidence was provided for the postulated mechanism involving buckling of a partially separated, thin slab. Spalls were usually lamellar, typically 10–20 grains in diameter and perhaps one to two grains thick. Grain-by-grain spallation produced by intense heat fluxes was not noticeable in this series of tests, but complex surface breakage, often generating several smaller spalls from a larger one, was quite common.

A separate experiment calibrated the torch heat flux to a flat surface for each of the conditions used during granite spallation experiments. Unfortunately, the time–temperature response of the surface of the copper block chosen for these tests was drastically different from that of spalling granite. The surface temperature of granite rises within less than 2 sec to about 500°C and remains somewhat constant thereafter once spallation initiates. The copper surface temperature never exceeded 250°C during any calibration runs, and response time of the system was several minutes. The high adiabatic flame temperature (over 2700°C), much hotter than either the granite or the copper, makes the effect of surface temperature on heat flux to the surfaces negligible.

Twelve thermocouples were positioned in a 38-cm dia disk by drilling thermowells to desired depths. The external circular surface was insulated with refractory bricks fitted to the disk's contour to restrict heat losses from the outer edge and the face opposite the heated area. The torch was directed horizontally such that the jet would impinge on the flat face opposite the thermocouple wells. Output from the reference junction was sent to a multichannel Hewlett-Packard voltage meter/scanner, which relayed voltages to a Hewlett-Packard 9835 desktop computer. Further experimental details are discussed by Rauenzahn [10].

Because this is an indirect method of determining the heat transfer rate to the copper and, by analogy, to the granite surface during the spallation experiments,

Table 3. Estimated spallation temperatures for Barre and Westerly granites under flame-jet heating and comparison of estimated impinging torch heat flux with literature results

Standoff	O_2 flow (m^3/sec STP)	Q_{max} (10^6 W/m^2)	r_0(m)	Experimentally estimated spallation temperatures (K) Barre	Westerly	Present study	Nusselt numbers (hD/k_{gas}) Hrycak	Popiel	Kataoka
8 in (20 cm)	3.2×10^{-3}	0.678	0.090	465	473	44	95	47	47
6 in (15 cm)	3.2×10^{-3}	0.890	0.079	435	446	58	92	49	58
8 in (20 cm)	5.1×10^{-3}	0.915	0.087	445	423	58	119	60	59
6 in (15 cm)	5.1×10^{-3}	1.19	0.075	472	455	76	114	60	74

Form of heat flux distribution: $Q = Q_{max} \exp(-r/r_0)$.
References [52, 56, 60].

further analysis is required. The data that exists as temperature–time histories for various cases must be transformed into heat fluxes at each point on the surface. The form of that heat flux distribution [$Q(r)$] was assumed, and the temperature field that would be induced by that heat flux on a copper block was computed by finite-difference techniques. Then, the parameters in the postulated heat flux expression were modified until a good match with the data was achieved.

Hrycak [52], Kumada and Mabuchi [53] and several others have suggested that heat transfer from a turbulent impinging jet to a flat surface decreases as local Reynolds number raised to a power:

$$Q(r) = Q_{max} Re_x^{-a'} = Q_{max}[(v\delta/\nu)_{bl}]^{-a'}, \quad (27)$$

where δ_{bl}, v and ν_{bl} are the boundary layer thickness, the free stream velocity at the edge of the shear layer and the local kinematic viscosity, respectively. Difficulties in accurately evaluating the parmeters in the Reynolds number, particularly when flame-jets are employed, and the singularity in $Re_x^{-a'}$ as $x \to 0$ make the exponential forms favoured by Anderson and Stresino [54] and Chamberlain [55] more tractable:

$$Q(r) = Q_{max} \exp(-(r/r_0)^{a''}). \quad (28)$$

Furthermore, measured radial variations in heat transfer and mass transfer from circular impinging jets in the works of many authors [52, 56, 57] suggest that a bell-shaped or Gaussian function, or even a simple decaying exponential ($a' = 1$), as proposed by Anderson and Stresino, might be appropriate for representing the heat flux distribution.

The major disadvantage of these simplified forms is the implicit assumption of constant heat flux at any radial position for the duration of the experiment. However, the adiabatic flame temperature for propane burning in oxygen is over 2700°C and the copper was never hotter than 300°C. The temperature difference that would be used in an appropriate convective heating source term $h(T_f - T_w)$ remains about 1200°C throughout the experiment, if we assumed a boundary layer temperature T_f about midway between the bulk gas and wall temperatures [58].

Optimum values of the three parameters in equation (28) were identified by minimizing an objective function based on the sums of the squares of the error between experimental and calculated temperatures:

$$\Phi = \sum_t \sum_{i=1}^{12} (T^t_{exp,i} - T^t_{calc,i})^2, \quad (29)$$

where the summation is performed over time for each experimental temperature at each recording scan of the HP analyzer. A rigorous method of function minimization, the simplex theory modified by Nelder and Mead [59], was adopted to automate the next choice of the three parameters throughout the optimization procedure.

The results of the heat flux analysis are shown in tabular form in Table 3. Because some preliminary searches located values for a' very close to 1, its value was assumed thereafter to be unity in the interest of increasing optimization speed. As a check, the maximum value of the heat flux was compared to that estimated from the cold temperature Nusselt number correlation of Hrycak for cooling provided by room-temperature air jets and those of Popiel *et al.* [56] and Kataoka *et al.* [60] for nonisothermal flame-jets impinging on flat surfaces. These independent sources of heat transfer values are also listed in the table, and the correlation of Kataoka *et al.* matches the current data extremely closely. Spallation temperatures estimated with a finite-difference heat conduction code using the surface heat flux distribution of measured spallation times are also displayed. As noted previously and as is evident from Table 3, the effect of applied heat flux on estimated spallation temperature is quite small, if any. However, given the scatter and limited range of the temperature data, this does not represent a stringent test of the Weibull theory.

LASER HEATING TESTS

The well-defined heating provided by CO_2 lasers (wavelength = 10.6 μm), as reported in similar previous investigations of thermally fracturing polycrystalline materials [25, 26], suggested a means for corroborating results gathered from the flame-jet tests. Spallation of granites under high heat fluxes provided by continuous wave lasers has been documented extensively in the past [26, 61, 62]. Earlier measurements indicated that over 90% of the incident beam energy was captured by the granitic surface [25, 26]. Under some operating conditions, the beam energy profile can be approximated by a simple Gaussian probability distribution [25]. There-

fore, in theory, complications introduced by measurement of the heat flux distribution, as in flame-jet heating, would be avoided by adopting laser heating for measuring surface spallation temperatures.

The chief disadvantage in employing lasers of less than 1 kW output is the small heated areas that result when using heat fluxes comparable to those expected from flame jets. Laser heating under these conditions might not produce spalls similar to those from granite blocks heated by a 30 mm dia flame-jet used in the field. For our tests, a Photon Sources continuous wave CO_2 laser with maximum power output of 500 W was available at MIT, but its nominal beam diameter was initially reported as only 5 mm, well below the typical characteristic spreading radius of the flame-jet described earlier. The distribution of energy within that diameter was unknown at the outset, and calibration of the heat source was again necessary.

Several blocks of Barre and Westerly granite, 10.2 × 10.2 × 3.8 cm, were sawed from the same stock used in the flame-jet tests. Although monitoring of radiative emission from the heated surface to measure surface temperature is currently being explored, high-speed videotaping of the experiment supplied measurement of spallation time. An experiment consisted of placing a block to be tested in the beam path while the shutter was closed, establishing the desired beam power, and starting the recording before opening the shutter. Power levels indicated by a meter measuring the laser output ranged from 62.5 to 400 W, inducing spallation at times between 0.1 and 8.0 sec. From a hand-held calorimeter placed in the beam path for 20 sec to capture the energy directed at it, the laser efficiency, or the amount of power actually supplied by the beam as a fraction of the meter reading, was determined to be 80% at 250 W (nominal) and 125 W (nominal). A glass lens with a focal length of 12.7 cm was situated in the beam path at 38.1 and 50.8 cm from the surface of the granite to induce larger (10 and 15 mm, nominally) beam diameters at incidence on the granite.

Violent release of chips from the underlying rock, characteristic of spallation by flame-jets, was not noticed in our laser tests. Usually, spalls would simply appear to peel away from the rock body, rather than be ejected. Partial melting of the biotitic mica in the surface layers impeded the chip's attempt to escape the surface, particularly during spallation of Barre granite. In some instances, single grains or portions of grains were seen to leap several cm from the surface, but at no time were large spalls with diameters greater than 10 spall thicknesses produced. The discrepancy between the spall sizes created during drilling and the spalls having a single grain characteristic of pinpoint heating suggests that these tests have not adequately reproduced spallation expected from flame-jet heating.

Measurement of the radial extent of the laser beam by burning Plexiglas revealed that spatial variation in incident heat flux was described best by a central core over which Q was constant ($r < r'$), followed by a nearly linear radial decrease until $Q = 0$ at $r = r''$. This is fairly consistent with the usually assumed Gaussian profile of beam energy content. However, extrapolating this to all power outputs is not justified. Lasers can alter firing modes as a function of power output. If beam energy profile varies with total contained power, subsequent analyses of temperature and stress fields are more complicated.

The experimental values of r' and r'' for two burn patterns were found to be strong functions of laser power output, and values for r' and r'' for other beam intensities were not estimated from the two known burn results. Likewise, use of the nominal beam diameter of 5 mm in subsequent thermal analyses led to significant errors in computed temperature and stress fields. Values of r' and r'' became increasingly difficult to measure accurately at high beam energies. Fortunately, from integration of the spatial distribution and an overall energy balance, we see that the maximum centreline heat flux (Q_{max}) is not extremely sensitive to small values of r':

$$Q_{max} = \frac{3}{\pi}\left(\frac{r'' - r'}{r''^3 - r'^3}\right) P_i. \tag{30}$$

Here, P_i denotes the total measured power contained in the beam (80% of the output meter reading). r'' has an appreciable impact on the resulting centreline heat flux, but its larger value was measured more accurately.

Conversion of local heat fluxes to a spallation temperature rise at the granite surface did not lend itself to analytical treatment, because the heating was not 1-D. The depth to which heat penetrated before spallation commenced was comparable to the width of the flat portion of the heat flux distribution. Thus, the TSAAS finite-element stress code used earlier as a method of examining stresses induced by heterogeneous rock properties was employed to compute temperature and stress distributions experienced by the granites.

As evident from Table 4, some of the spallation temperatures are considerably higher than those from the propane torch spallation experiments given in Table 3. They are also higher than any previously mentioned literature values. Furthermore, the disagreement between experiments conducted with different beam conditions is extreme. Uncertainties in the exact outermost extent of the beam could easily account for the unusually high computed temperatures. More importantly, however, the localized heating to which the rock was subjected invalidates the assumption of 1-D heat conduction. In 1 sec, thermal effects will penetrate approx. 1 mm in granite, which is not much smaller than the size of the energy-containing portion of the beam

Table 4. Estimated spallation temperatures from laser experiments

Power (W)	Centreline temperature (K)
	Westerly granite
100	520
200	1120
	Barre Granite
100	710

(1–2 mm). As a consequence, the simple relation between temperature and stress fields given by Gray [13], which formed the basis for the extension of Weibull's theory to spallation, no longer holds. The spallation temperature concept proposed earlier, which appears to be verified by the propane torch experiments, relies heavily on this easily defined correspondence of temperature and stress fields. Therefore, rock surface temperatures under the centreline will not agree with estimates from equation (25). If Weibull's theory holds,

$$\int_V (\sigma/\sigma_0)^m \, dV \; (\equiv I_\sigma)$$

must be calculated for situations in which stress and temperature differences are not easily computed analytically.

The stress integral (I_σ) was computed from the stress field predicted by the finite-element code TSAAS. Once again, the relevant volume of rock over which I_σ was computed is a cone with its apex at the centreline and a radius of C_{Lx}, where x is the axial depth into the rock. I_σ should be about 0.7 for reasonable values of cumulative failure probability G [$G(\sigma) = 1 - \exp(-I_\sigma)$]. In this case, σ_0 and m are both, in principle, unknown, though Dey and Kranz [52] and Lundborg [49] report m to be between 10 and 25 for the granites they tested. Unfortunately, I_σ is extremely sensitive to the exact value of m chosen. For the experimental conditions with measured beam profiles, given values of σ_0 of 50 and 70 MPa-(m)$^{3/m}$, values of m that are consistent with numerically estimated spallation temperatures are between 13 and 25.

COMPARISON OF EXPERIMENTAL RESULTS AND THEORETICALLY PREDICTED TEMPERATURES

For the pseudo-1-D heating supplied by the propane torch, equation (26) relates the applied heat flux to the predicted spallation surface temperature rise of the rock as given by Weibull's statistical failure theory. Thermal and elastic rock property values, even at room temperature, are uncertain, often to within a factor of two, and the effect on the predicted values can be substantial. For example, Young's modulus (E) for Barre granite, here taken as 45 GPa, could easily vary by 50%, depending on the sample chosen, the testing conditions, the direction of the measurement relative to the bedding plane of the quarry, or even location within the quarry [14]. This relatively small variation in E will yield almost 50% uncertainty in spallation temperature, potentially increasing it from 400 to 600°C.

More realistically, Weibull theory can be examined for its ability to predict spallation temperatures in the correct range for typical temperature-averaged property values of Barre and Westerly granites. As displayed in Fig. 9, the magnitude of spallation temperatures computed from the finite-difference numerical scheme and from Weibull's theory are reasonable but show no significant sensitivity to applied heat flux. At first glance, the agreement between theory and experiment seems poor. However, the predicted temperatures are somewhat dependent on the exact value of Weibull's homogeneity parameter (m). In fact, if probable ranges of m (15–25) are substituted in equation (26), the range of theoretical temperatures easily encompasses the scatter in experimental spallation temperatures, as shown in Fig. 10. This holds even if maximum probable errors in estimating spallation times and heat fluxes are included. Consequently, a more sensitive method of testing the applicability of the Weibull theory is required.

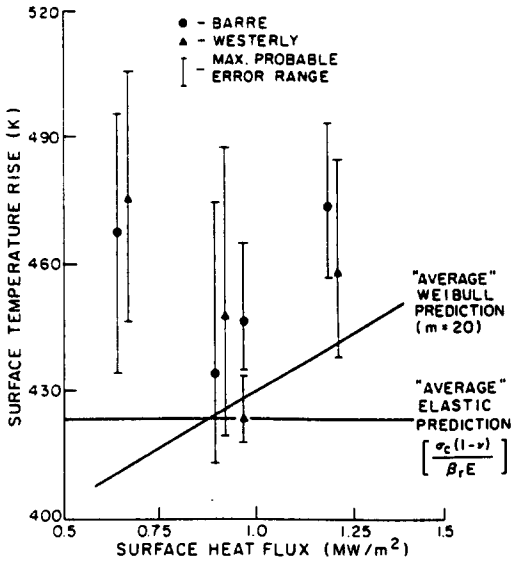


Fig. 9. Experimental rock surface spallation temperature rise as a function of applied heat flux, as determined from copper calorimetry tests. Also shown are average Weibull and simple thermoelastic theoretical predictions.

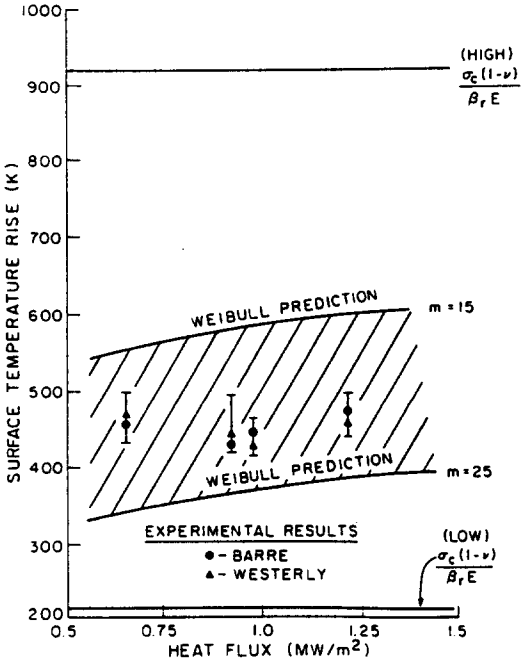


Fig. 10. Experimental rock surface spallation temperature rise as a function of applied heat flux with reasonable theoretical bounds.

The simple stress model first proposed by Preston [12] and later by Gray [13] may be reasonable for spallation on a macroscopic scale. A comparison of the experimental spallation temperature rise with that calculated from the simple thermoelastic model:

$$\Delta T_s = \frac{\sigma_c(1-\nu)}{\beta_t E}, \tag{31}$$

shows remarkable agreement, even assuming room-temperature properties. The effect of variation in physical parameters is considerable (Fig. 10), as displayed by computing the maximum (high) and minimum (low) probable values of the right-hand side of equation (31). Because the width of predicted ΔT_s covers the experimental points, the error introduced by using room temperature properties may not be any larger than if more refined values were employed. The more apparent disadvantage of this model is its reliance on such a simple mechanism and mathematical description of rock failure.

As a side issue, the importance of quartz thermomechanical and structural properties in characterizing spallation quality of rock should be reviewed. A common argument for the relatively excellent spallability of quartzitic species has been the effect of the rapid increase in thermal expansion coefficient of quartz near the α–β phase transition temperature of 573°C [28]. As more evidence against such a claim, the results of this spallation temperature study add to previous data of spallation temperatures that are all below this equilibrium condition. In fact, none of the temperatures calculated from the propane experiments were above 573°C, even with maximum levels of uncertainty included.

Spallation temperatures computed from the laser tests do not carry as much significance as those produced under propane torch heating. The unusually high temperatures obtained for the highest laser power with the most focused beams cannot be viewed with much confidence for several reasons. Uncertainties about the accuracy of the Plexiglass burn method for measuring Q_{max} can introduce serious errors when estimating surface temperatures. Most importantly, the heating characteristics of the laser often caused single grains or even fractions of single grains to be most severely affected. Stresses present under more spatially uniform heat fluxes can sample a more representative volume of near surface layers to choose the most critical flaw available. Thus, if thermal failure of rock is induced by extremely localized heating, the material could exhibit a remarkably high apparent strength, not only because a much smaller rock volume is stressed but also because the thermal stress field did not interact with a statistically typical distribution of flaws.

CONCLUSIONS

This paper has reported surprising consistency between experimentally estimates spallation surface conditions, particularly temperatures indirectly measured with propane torch spallation, and those predicted by using a fairly simple statistical failure model. This failure description does not rely on a careful description of grain–grain micromechanics. Rather, it merely uses a probabilistic distribution of failure strengths due to Weibull based on the notion that sample destruction depends only on the weakest flaw encountered by the stress field. Thus, if the appropriate Weibull parameters are available, spallation temperatures and expected penetration rates are easily determined.

Despite these advantages and agreement with available experimental data, the use of Weibull statistics for describing the mechanics of rock spallation is hampered by several major difficulties. General applicability of Weibull statistical failure theory to compressive failure, even when only a few flaws are involved, is not firmly established. Weibull statistics fit the tensile failure size–strength data extremely well, especially for such isotropic materials as glasses. Prior to now, few researchers [48–50] have ventured to suppose that Weibull statistics could be useful in cases where compression dominates. Future stringent testing of Weibull theory for spallation and other instances in which compressive stresses act on a very small scale will probably support its use in these situations, but the question remains open for the present.

In instances where rock is rapidly heated, generation of microcracks during the heating process is well-documented and can be explained through simple models of grain–grain and grain–heterogeneity interactions, as shown previously. Omission of this behaviour from the rock failure model proposed here may introduce errors. The simplest manner in which to compensate for this observed behaviour would be to introduce measured temperature-dependent Weibull parameters. These measurements would be tedious and difficult to accomplish in practice.

By contrast, the alternative of adopting a more deterministic failure model is also unattractive. The wide variety of rock types and their origins would necessarily require individual examination and unique model development for each rock variety. The chance of systematically synthesizing an all-inclusive thermomechanical model of rock spallation is accordingly small, and improvements in measurement techniques and theoretical development of global, lumped-parameter phenomenological descriptions are more realistic. Thus, extensions of Weibull theory proposed by us and Dey [50] and expanded in this research provide a method of including important rock structural information not considered by simpler models [13] without development of extremely specific descriptions of a rock's individual morphology.

Until further testing of the Weibull theory is available, the temperature of rock spallation can be, for engineering purposes, estimated from equation (14). If values of the Weibull parameters are unavailable or uncertain, then ΔT_s can be more crudely estimated from equation (31). Likewise, the susceptibility of a certain species to penetration can be roughly determined, if its spalls at all, from a criterion directly derived from the heat balance

at the rock–fluid interface:

$$\pi = \frac{\beta_r E}{\sigma_c (\rho C_p)_r} \approx \frac{V_{dr}}{Q}. \quad (32)$$

Finally, a simplified version of the crtierion derived by Kolodko [24] should prove useful in assessing the spallability of certain rock samples. Then, only its relative spallability needs to be quantified [equation (32)] by property measurements or field drilling tests.

In the future, by using radiometric methods, we plan to perform continuous, direct measurements of spallation surface temperatures produced by flame-jets. Our work will also use higher power, larger beam lasers to eliminate uncertainties introduced by small beam areas. Through these more refined tests of Weibull theory, we expect to answer some of the remaining questions related to spallation mechanics.

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